

Total No. of Questions : 8]

SEAT No. :

PE-2488

[Total No. of Pages : 2

[6583]-12

T.E. (Civil)

**DESIGN OF STEEL STRUCTURES
(2019 Pattern) (Semester - V) (301003)**

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.
- 5) IS:800-2007 & IS:875 part I to III are permitted.

- Q1)** a) Design moment resisting base for a ISHB 250 @54.7 kg/m column to carrying a factored load of 650 kN and factored bending moment 50 kN-m. [12]
b) Which of the two buckling or stiffness of compression members is more critical? [5]

OR

- Q2)** a) Design a column base for the column consisting of ISWB300 @ 48.1 kg/m carrying axial load 740 kN. Design welded connection between column and base plate. The column is to be supported on concrete pedestal to be built with M20 grade. [12]
b) In what sense column caps are similar to column base plates? [5]

- Q3)** a) A simply supported beam of 5 m effective span. It carries a uniformly distributed load w kN/m throughout the span including self-weight. The compression flange of beam is laterally unsupported throughout the span. Determine the intensity of udl of w , so that the section ISMB 400 @ 61.6 kg/m use for beam can carry safely. [10]
b) A simply supported beam of 4 m effective span. It carries a uniformly distributed load 60 kN/m throughout the span including self-weight. Design the beam if the compression flange is laterally restrained throughout the span. [8]

OR

- Q4)** a) A simply supported beam spanning 8 m carries a uniformly distributed load 5 kN/m throughout the span including self-weight. The compression flange of beam is laterally supported throughout the span. Design section and check for serviceability. [13]
b) How are the column buckling and the lateral buckling of beam similar? [5]

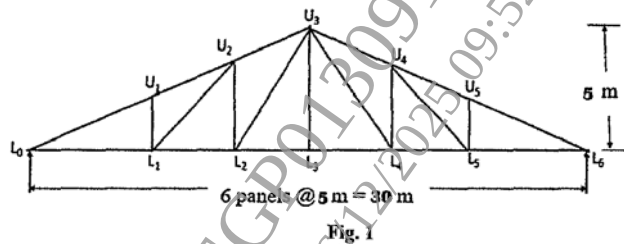
P.T.O.

Q5) Design a gantry girder supporting an electric overhead travelling crane to the following data: [17]

- Capacity of crane = 280 kN
- Span between crane rails = 14 m, Self-weight crane girder = 150 kN
- Weight of crab, electric motor, Hook etc. = 15 kN
- Minimum hook approach = 1.0m
- Wheelbase = 3.5m
- Span of Gantry 7.0 m
- Self-weight of trolley = 105kN
- Self-weight of rails = 0.3 kN/m
- Height of rail section = 75 mm

OR

Q6) Determine the design force in the member L_0U_1 , L_0L_1 , and U_1L_1 of pratt truss as shown in Fig. 1. Assume design wind pressure is 1200 N/m^2 , use A.C. Sheet and the C/C spacing of truss is 6m. Assume self-weight of purlin 120 N/m . [17]



Q7) A simply supported welded plate girder of an effective span of 26 m subjected to uniformly distributed load 40 kN/m throughout the span excluding the self-weight of plate girder. Assume compression flange laterally supported throughout the span and yield stress of steel is 250 MPa . Design cross section of plate girder, bearing and intermediate stiffeners. Draw sectional plan and elevation. [18]

OR

Q8) Design a suitable cross section of welded plate girder 30 m in span and laterally supported throughout. It has to support a uniform load of 30 kN/m throughout the span including of self-weight of plate girder. It is also loaded with two concentrated load of 140 kN acting at 10 m from either supports. Design C/S of plate girder without intermediate stiffeners. The steel for the flange and web plates is of grade Fe 410. Yield stress of steel may be assumed to be 250 MPa irrespective of the thickness of plates used. [18]

